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In [41]: from nltk.util import ngrams
from collections import Counter
import nltk
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
import numpy as np

nltk.download('punkt_tab')
unigram_freq_khadi_boli = 0
unigram_freq_hindi = 0

with open("Khadi_Boli_Corpus.txt", 'r') as file:
    khadi_boli_corpus = file.read().lower()
    tokens = nltk.word_tokenize(khadi_boli_corpus)

    unigrams = list(ngrams(tokens, 1))
    unigram_freq_khadi_boli = Counter(unigrams)

with open("Hindi_Corpus.txt", 'r') as file:
    hindi_corpus = file.read().lower()
    tokens = nltk.word_tokenize(hindi_corpus)

    unigrams = list(ngrams(tokens, 1))
    unigram_freq_hindi = Counter(unigrams)

all_unigrams = set(unigram_freq_khadi_boli.keys()).intersection(set(unigram_freq_hindi.keys()))

data = []

for bg in all_unigrams:
    data.append([
        " ".join(bg),
        unigram_freq_khadi_boli.get(bg, 0),
        unigram_freq_hindi.get(bg, 0)
    ])

df = pd.DataFrame(data, columns=["Unigram", "Khadi Boli", "Hindi"])

print(df)

df_plot = df.set_index("Unigram")

df_plot.plot(kind='bar', figsize=(12,6))

plt.title("Unigram Frequency Comparison")
plt.xlabel("Unigrams")
plt.ylabel("Frequency")

plt.xticks(rotation=45)

plt.tight_layout()
plt.show()

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plt.figure(figsize=(8,6))

plt.scatter(df["Khadi Boli"], df["Hindi"])

for i, txt in enumerate(df["Unigram"]):
    plt.annotate(txt,
                  (df["Khadi Boli"][i],
                   df["Hindi"][i]))

plt.xlabel("Khadi Boli Frequency")
plt.ylabel("Hindi Frequency")
plt.title("Scatter Plot of Unigram Frequencies")

plt.grid(True)

plt.show()

X = df["Khadi Boli"].values.reshape(-1,1)
y = df["Hindi"].values

model = LinearRegression()
model.fit(X, y)

y_pred = model.predict(X)

plt.figure(figsize=(8,6))

sns.scatterplot(x=df["Khadi Boli"],
                 y=df["Hindi"],
                 s=100)

plt.plot(df["Khadi Boli"],
         y_pred)

plt.xlabel("Khadi Boli Frequency")
plt.ylabel("Hindi Frequency")

plt.title("Linear Regression Between Dialects")

plt.grid(True)

plt.show()

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[nltk_data] Downloading package punkt_tab to
[nltk_data] /home/0960eef3-c09c-4e5d-bcb4-
[nltk_data] 78b159a319bd/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

```

	Unigram	Khadi Boli	Hindi
0	कहाँ	1	4
1	आ	1	1
2	मैं	3	3
3	खाया	1	2
4	जा	2	2
5	है	5	9
6	खाओगे	1	1
7	बात	1	1
8	के	9	1
9	हो	1	5
10	बोल	1	1
11	था	1	3
12	गया	1	1
13	हूँ	2	3
14	आया	2	2
15	देख	1	1
16	जाना	1	1

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/tmp/ipykernel_233/4264468972.py:56: UserWarning: Glyph 2325 (\N{DEVANAGARI LETTE
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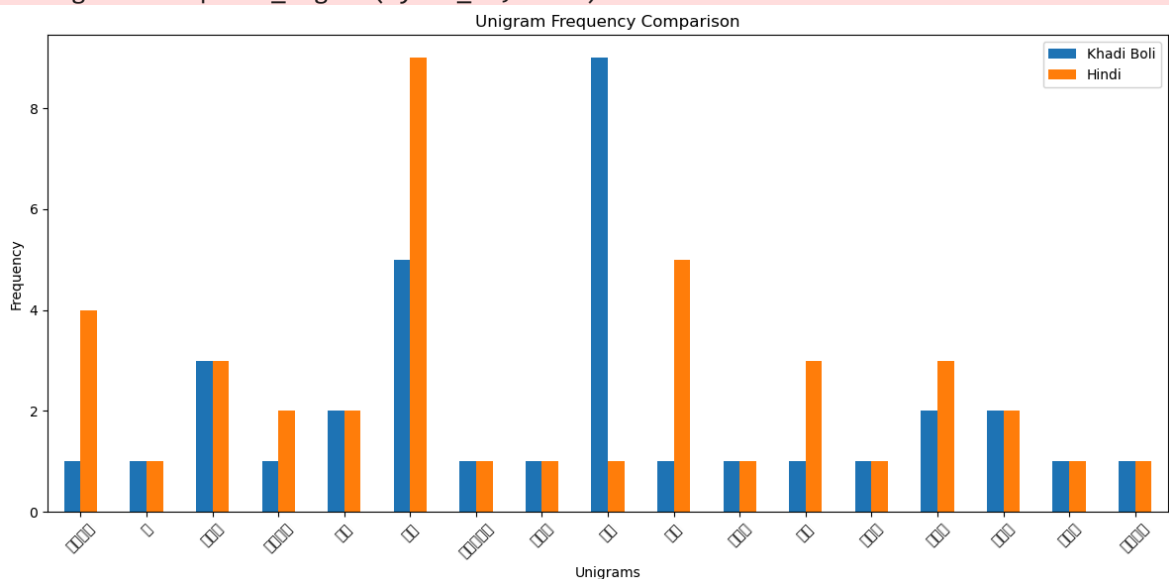
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/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2306 (\N{DEVANAGARI SIGN ANUSVARA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2310 (\N{DEVANAGARI LETTER AA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2350 (\N{DEVANAGARI LETTER MA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2376 (\N{DEVANAGARI VOWEL SIGN AI}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2326 (\N{DEVANAGARI LETTER KHA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2351 (\N{DEVANAGARI LETTER YA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2332 (\N{DEVANAGARI LETTER JA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2323 (\N{DEVANAGARI LETTER O}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2327 (\N{DEVANAGARI LETTER GA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2375 (\N{DEVANAGARI VOWEL SIGN E}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2348 (\N{DEVANAGARI LETTER BA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2340 (\N{DEVANAGARI LETTER TA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2379 (\N{DEVANAGARI VOWEL SIGN O}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2354 (\N{DEVANAGARI LETTER LA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2341 (\N{DEVANAGARI LETTER THA}) missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)

```

```
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2370 (\N{DEVANAGARI VOWEL SIGN UU}) missing from font(s) DejaVu Sans.
```

```
fig.canvas.print_figure(bytes_io, **kw)
```

```
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2305 (\N{DEVANAGARI SIGN CANDRABINDU}) missing from font(s) DejaVu Sans.
```

```
fig.canvas.print_figure(bytes_io, **kw)
```

```
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2342 (\N{DEVANAGARI LETTER DA}) missing from font(s) DejaVu Sans.
```

```
fig.canvas.print_figure(bytes_io, **kw)
```

```
/opt/conda/envs/anaconda-ai-2025.12-py312/lib/python3.12/site-packages/IPython/core/pylabtools.py:170: UserWarning: Glyph 2344 (\N{DEVANAGARI LETTER NA}) missing from font(s) DejaVu Sans.
```

```
fig.canvas.print_figure(bytes_io, **kw)
```

